Bad Controls in Dynamic Panels: A Diagnostic Formula and a Benchmark Specification for the Contemporaneous Treatment Effect

Manoel Galdino* Davi Moreira† Carolina Dolleans‡

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Abstract

We study a practical problem in time-series cross-sectional (TSCS) research: whether observed time-varying covariates should be included as controls when the estimand is the contemporaneous treatment effect (CET). The same covariate can function as a confounder, a collider, or a mediator depending on its timing relative to the treatment, and choosing wrong can introduce bias rather than remove it. We present the Included Variable Bias (IVB) formula—IVB $= -\theta^\star \times \pi$—a closed-form diagnostic, based on the well-known short-versus-long regression decomposition, that quantifies the change in the treatment estimate caused by including or excluding a covariate, using quantities the researcher already has. The formula mirrors the classic Omitted Variable Bias formula but addresses the opposite problem; combined with a Directed Acyclic Graph, it tells the researcher whether the change represents collider bias, omitted variable bias, or over-control of a mediator. We show that in TSCS settings, a benchmark ADL + FE specification with lagged state variables avoids conditioning on contemporaneous post-treatment covariates: it blocks collider paths, preserves the total contemporaneous effect when covariates are lagged rather than contemporaneous, and absorbs confounding through lagged values. Monte Carlo simulations covering collider, dual-role, and mediator cases across 238 DGP configurations confirm that this benchmark keeps bias below 3% of the treatment effect. We apply the diagnostic to six published studies spanning democratization, peacekeeping, fiscal capacity, infrastructure, and sovereign debt, classifying each control via DAG-based reasoning. Among 14 theoretically motivated collider candidates, the median IVB is approximately 0.13 standard errors—well within sampling noise.

*Assistant Professor of Political Science, Universidade de São Paulo. E-mail: mgaldino@usp.br
†Professor of Quantitative Methods, Mitch Daniels School of Business, Purdue University.
‡Graduate Student of Political Science, Universidade Federal de Pernambuco.

1 Introduction

Every researcher who includes a time-varying covariate in a panel regression is making a bet: that the variable removes more confounding than it introduces in distortion. The bet is often wrong. If the covariate is a collider—caused by both the treatment and the outcome—including it opens a spurious path and biases the estimate. If it is a contemporaneous mediator, including it blocks the indirect causal channel and attenuates the total effect. And in TSCS data, the same variable can switch roles depending on when it is measured: GDP per capita at $t-1$ may be a confounder that the researcher needs; GDP per capita at t may be a collider or mediator that the researcher should avoid. Standard heuristics for control selection—include causes of the outcome, or common causes of treatment and outcome—do not distinguish these cases (Cinelli, Forney and Pearl, 2022).

The problem is recognized on both sides of a disciplinary divide that has produced parallel but disconnected responses. In the DID literature, Caetano and Callaway (2024) and Lin and Zhang (2022) have shown that time-varying covariates create specification problems even under parallel trends, and the "bad controls'' problem in DID is the subject of active research (Caetano et al., 2022). In the TSCS literature, Blackwell and Glynn (2018) and Imai and Kim (2021) have recommended ADL models that condition on lagged states. Neither tradition offers an estimable formula that tells the applied researcher how much a specific covariate inclusion changes the treatment estimate.

This paper provides such a formula. The Included Variable Bias (IVB) diagnostic repurposes the standard short-versus-long regression decomposition:

$$\text{IVB} = -\theta^\star \times \pi \tag{1}$$

where $\theta^\star$ is the coefficient on the covariate in the augmented regression and π is the coefficient on the treatment in an auxiliary regression of the covariate on the treatment and legitimate controls.[1] Both components come from regressions the researcher has already run or can run immediately. The formula mirrors the Omitted Variable Bias formula but works in the opposite direction: where OVB measures the cost of leaving out a confounder, IVB measures the change from putting in an additional variable. The algebra is an identity of OLS via the Frisch–Waugh–Lovell theorem (Frisch and Waugh, 1933; Lovell, 1963); it holds in cross-sectional, TWFE, and ADL models for any treatment type. What makes it a diagnostic, not just a decomposition, is the DAG: the graph tells the researcher whether the change is collider bias, omitted

[1] In the cross-sectional derivation (Section~4.1), $\theta^\star$ corresponds to $\beta_2^\star$ and π corresponds to ϕ_1; the cross-sectional notation appears in that section and its comparison table, while we use the TSCS notation ($\theta^\star$, π) in all subsequent discussion.

variable bias, or over-control of a mediator.

The formula is the first of two tools. The second is a benchmark specification. For the contemporaneous treatment effect (CET) in TSCS settings, ADL + FE with lagged state variables avoids the problems that contemporaneous covariates create. The lagged value Z_{t-1} absorbs confounding without opening a collider path or blocking the mediator channel $D_t \to Z_t \to Y_t$. This timing logic exploits a structural advantage of TSCS data over canonical two-period DID, where the researcher cannot separate the confounding role of Z from its post-treatment role. Monte Carlo simulations across 238 DGP configurations—spanning collider, dual-role, and mediator cases with nonlinearities and outcome-to-treatment feedback—confirm that the ADL benchmark keeps bias below 3% of the treatment effect.

We apply the diagnostic to six published studies in comparative politics and international political economy. For each study, we classify every control variable using DAG-based reasoning grounded in the published literature. Among 14 theoretically motivated collider candidates, the median |IVB/SE| is approximately 0.13—about one-eighth of a standard error. Only GDP per capita in a growth regression (Rogowski et al., 2022) exceeds one standard error, and its collider status is causally ambiguous because it is simultaneously a confounder. The typical finding is that the specification choice does not matter much. Knowing that is itself the point: the formula lets the researcher verify, rather than assume, that including or excluding a covariate is quantitatively inconsequential.

The remainder of this paper is organized as follows. Section~2 reviews the standard heuristics for selecting control variables and identifies their limitations. Section~3 introduces DAGs, illustrates collider bias in a TSCS context, and shows how timing determines a covariate's causal role. Section~4 presents the IVB formula for cross-sectional, TWFE, and ADL models. Section~5 uses Monte Carlo simulations to benchmark dynamic specifications for the CET across collider, mediator, and dual-role scenarios. Section~6 applies the formula to six published studies. Section~7 concludes with a practical rule, explicit limits, and directions for future work.

2 The Control Variable Problem

Two widely used heuristics guide control selection in applied work. *Control checking* includes causes of the outcome to improve precision (e.g., Dietrich, 2016); *confounding checking* targets joint causes of treatment and outcome to block omitted variable bias (e.g., Hegre et al., 2001). Neither asks whether a candidate control is a common *effect* of treatment and outcome—a collider whose inclusion opens rather than closes backdoor paths (Cinelli, Forney and Pearl, 2022)—nor whether it is a mediator whose inclusion blocks the

indirect causal channel. In TSCS settings, this blind spot is compounded by the temporal structure of the data: a variable that is a confounder when measured before treatment ($Z_{t-1} \to D_t$) may become a collider or a mediator when measured contemporaneously ($D_t \to Z_t$).

Even formal approaches leave a quantification gap. The DAG framework (Pearl, 2009), which we employ here, tells researchers *whether* a variable is a collider but not *how much* bias results from conditioning on it. Prior work has quantified collider bias in specific configurations—collider-stratification bias (Greenland, 2003), M-bias in LSEMs (Ding and Miratrix, 2015), path-analytic expressions (Pearl, 2013), and disparate impact (Jung et al., 2024)—but these formulas are parameterized in structural LSEM coefficients that are typically unobserved, and none addresses ADL models or TSCS settings.[2]

Our IVB formula fills this gap by expressing the change from including (or excluding) a covariate in terms of regression coefficients the researcher already has ($\hat{\theta}^\star$ from the long regression, $\hat{\pi}$ from a simple auxiliary regression)—analogous to how Cinelli and Hazlett (2020) recast OVB sensitivity in terms of partial R^2. The practical asymmetry is notable: whereas OVB requires observing the omitted confounder, IVB uses quantities already estimated since the covariate is included by assumption. Combined with a DAG, the formula tells the researcher whether the arithmetic difference between specifications represents collider bias (exclude the variable), omitted variable bias (keep the variable), or over-control of a mediator (keep only lagged values). The formula extends naturally to ADL models commonly used in TSCS analysis. Because the IVB is an algebraic identity of OLS, it holds exactly for any treatment type without separate derivations (Ding and Miratrix, 2015).

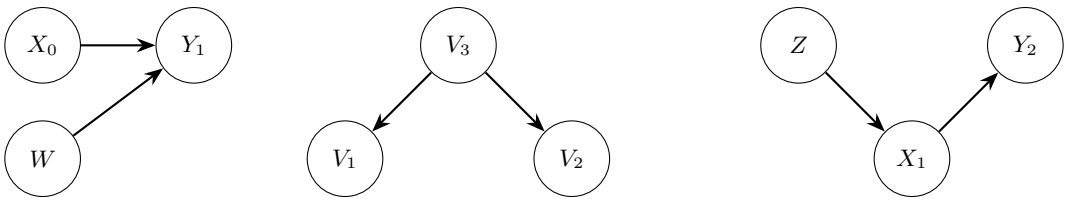
3 DAGs, Timing, and Causal Roles

3.1 Preliminaries

Directed Acyclic Graphs (DAGs) are part of an approach to causal inference first developed by Wright (1928) and Wright (1934), and substantially advanced by the work of Pearl (2009) and collaborators (Pearl and Mackenzie, 2018). DAGs have been applied in several domains, including epidemiology, computer science, and, more recently, political science (Imbens, 2020).

All DAGs comprise three primary structures: chains, forks, and colliders (or inverted forks), shown in Figure~1. Understanding these structures is essential for correct control variable selection.

[2]Ding and Miratrix (2015) showed that pure M-Bias tends to be a higher-order effect—a product of four structural coefficients—that is typically small unless all four are simultaneously large. They further showed that conditioning on a variable that is simultaneously collider and confounder reduces bias in roughly 75% of the feasible parameter space. Jung et al. (2024) used the exact term "included variable bias'' in the context of disparate impact estimation.



(a) Collider **(b) Fork** **(c) Chain**

Figure 1: The three elementary structures of DAGs: (a) a collider or inverted fork, where Y_1 is a common effect of X_0 and W; (b) a fork, where V_3 is a common cause of V_1 and V_2; and (c) a chain, where X_1 is a mediator between Z and Y_2.

In a **fork** (panel b), V_3 is a common cause of V_1 and V_2. If one is interested in the effect of V_1 on V_2, controlling for the common cause V_3 blocks the backdoor path and eliminates omitted variable bias. In a **chain** (panel c), X_1 mediates the effect of Z on Y_2. If one is interested in the total effect, controlling for the mediator blocks the causal path and leads to underestimation. In a **collider** (panel a), Y_1 is a common effect of X_0 and W. A collider, unlike chains and forks, does not induce an association between its parent variables. However, conditioning on a collider (or on its descendants) *creates* a spurious association between its causes (Elwert and Winship, 2014; Pearl and Mackenzie, 2018).

The heuristic behind "confounding checking'' can be understood as an attempt to block all backdoor paths by conditioning on forks. However, this heuristic says nothing about what happens when the candidate control is a collider. This is the problem we now illustrate in a TSCS context.

3.2 A TSCS Application: Political Change and Civil War

Consider a typical setup in comparative politics. A researcher wants to estimate the causal effect of political regime change on civil war, as in Hegre et al. (2001). Suppose the only measured confounder is per capita income, which plausibly causes both political change and civil war (Blattman and Miguel, 2010). The researcher controls for per capita income to block the backdoor path.

In a TSCS setting, variables must be indexed by time. The researcher decides to lag both per capita income and political change to address reverse causality concerns. The resulting model corresponds to the temporal DAG in Figure~2.

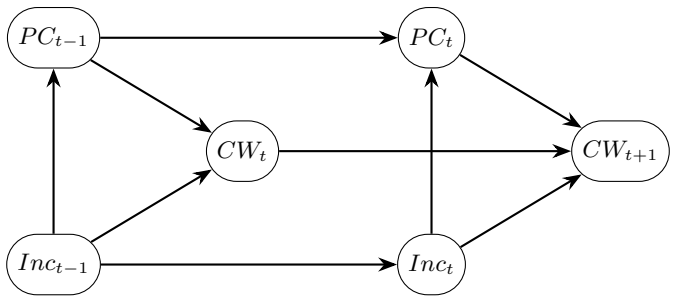


Figure 2: Temporal DAG for the effect of Political Change (PC) on Civil War (CW), with Per Capita Income (Inc) as a confounder. The temporal structure reveals that a lagged dependent variable (CW_t) is needed to block all backdoor paths to CW_{t+1}.

A key insight from this DAG is that the dynamic structure introduces backdoor paths through the lagged outcome (CW_t). To block these paths, the researcher must include CW_t as a control—leading to a dynamic panel (ADL) specification. Without the lagged dependent variable, the causal effect is biased.

3.3 Foreign Collider Bias

Now suppose the researcher, following the confounding checking heuristic, wants to add Democracy Level (Dem) as an additional control, as Hegre et al. (2001) suggest. At first glance, this seems harmless: democracy plausibly affects both political change and civil war. However, a more careful DAG reveals a critical problem.

It is plausible that civil war *causes* changes in democracy levels (countries that experience civil war tend to become less democratic). If, in addition, there exist unobserved factors U that affect both democracy levels and civil war—such as institutional quality, state capacity, or regional instability—then democracy level becomes a **collider**: it is a common effect of civil war and the unobserved U. Conditioning on it opens a backdoor path connecting U to civil war through democracy level. Figure~3 displays this structure.

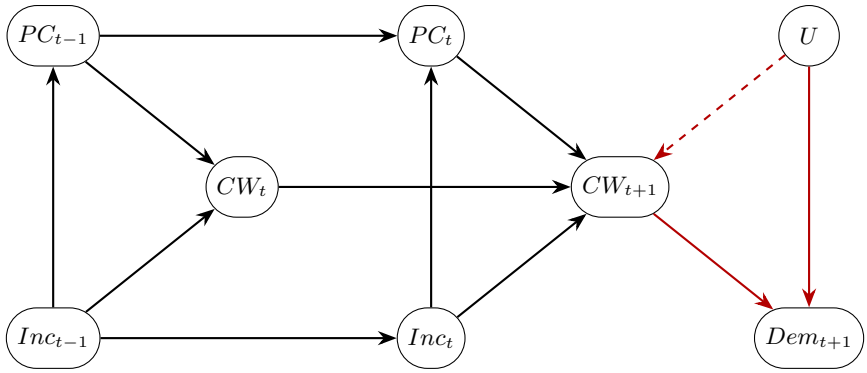


Figure 3: Complete DAG showing Democracy Level (Dem_{t+1}) as a collider. Civil War and an unobserved variable U both cause Democracy Level. Conditioning on Dem_{t+1} opens the collider $CW_{t+1} \rightarrow Dem_{t+1} \leftarrow U$, creating a spurious association between CW_{t+1} and U. Because U also causes CW_{t+1} (dashed arrow), this biases the estimate of the effect of Political Change on Civil War. Red arrows highlight the collider structure.

We use the term *foreign collider bias* descriptively, not analytically—it denotes the practical difficulty that the causal knowledge needed to identify a collider often resides in literatures outside the researcher's primary domain. In this example, discovering that Democracy Level is a collider requires consulting the `foreign'' literature on the determinants of democracy, which is typically separate from the civil war literature motivating the original study. The phenomenon is not limited to economic variables: in studies of voter turnout (Y) and campaign spending (D), a researcher controlling for media coverage (Z) faces the same pattern if both spending and turnout causally affect media coverage---a relationship documented in the communications literature rather than the electoral behavior literature. The standard heuristics for including controls review the literature on the \textit{outcome} and the \textit{treatment}, but they do not instruct the researcher to investigate whether the candidate control is itself caused by the treatment or outcome through an unobserved variable. The underlying causal phenomenon is standard collider bias; what`foreign'' adds is a practical warning about the disciplinary boundaries that make such colliders easy to miss.

The concept of foreign collider bias has broad applicability in TSCS research. Consider a researcher studying the effect of democratization on ethnic inequality who controls for GDP per capita (Leipziger, 2024). The `domestic'' literature on ethnic inequality identifies GDP as a correlate of inequality and a plausible confounder. But the`foreign'' literature on economic growth establishes that democracy causes GDP growth (Acemoglu et al., 2019), and that ethnic inequality causally reduces GDP (Grundler and Link, 2024). These two causal channels—$D \rightarrow Z$ and $Y \rightarrow Z$—make GDP per capita a collider as well as a confounder. Discovering this requires consulting the growth literature, not just the

inequality literature. Similarly, in studies of public infrastructure and economic growth (Rogowski et al., 2022), GDP per capita is simultaneously the confounding variable that growth regressions must control for (conditional convergence) and a collider through the paper's core mechanism ($D \to Z$). This pattern is particularly easy to miss because researchers following best practices—reviewing the literature on potential confounders—may not think to investigate the determinants of each candidate control variable in adjacent literatures.

This example illustrates the power of DAGs: by making the causal structure explicit, they reveal that Democracy Level is a collider and that conditioning on it biases the regression. But the DAG alone does not tell us *how much* bias results. For that, we need a formula.

3.4 Timing Determines Causal Role

The civil war example illustrates collider bias, but the same variable can play different causal roles depending on its temporal index. This is the central complication of control variable selection in TSCS settings: the researcher must decide not only *which* variables to include, but *when* they are measured. We distinguish three cases that are central to the analysis that follows.

Collider. When both the treatment and the outcome (or an unobserved common cause of the outcome) causally affect Z within the same period—$D_t \to Z_t \leftarrow Y_t$—conditioning on Z_t opens a spurious path. This is the foreign collider bias case illustrated above.

Contemporaneous mediator. When the treatment affects Z and Z in turn affects the outcome within the same period—$D_t \to Z_t \to Y_t$—the variable Z_t is a mediator. Conditioning on Z_t blocks the indirect causal channel $D_t \to Z_t \to Y_t$, so the regression recovers only the direct effect of D_t on Y_t. If the researcher's estimand is the *total* contemporaneous effect (CET), including Z_t introduces over-control bias: the estimate is attenuated by exactly the indirect effect that Z_t mediates.

Dual role across time. In dynamic settings, the same variable often plays *both* roles across different time indices. The lagged value Z_{t-1} may be a confounder—affecting both D_t and Y_t through persistence—while the contemporaneous value Z_t is a collider or mediator. This dual role creates a tension: excluding Z entirely leaves confounding from Z_{t-1} unaddressed, while including Z_t introduces collider or mediator bias. The resolution is to condition on Z_{t-1} but not on Z_t: the lagged value absorbs the confounding channel without opening the contemporaneous collider path or blocking the contemporaneous mediator path. This is precisely the logic behind the ADL specification with lagged state variables that we benchmark in Section~5.

The IVB formula developed in the next section quantifies the arithmetic difference between specifications that

include and exclude a covariate. Whether that difference represents collider bias, over-control, or appropriate deconfounding depends on the DAG. But in all three cases, the formula gives the researcher a number—not just a qualitative classification—making the specification choice transparent and auditable.

4 The Included Variable Bias Formula

In this section, we present the Included Variable Bias (IVB) formula—a closed-form expression for the change in the treatment effect estimate caused by including an additional covariate. The decomposition is a direct application of the Frisch–Waugh–Lovell theorem and the standard short-versus-long regression identity (see, e.g., Angrist and Pischke, 2009, Theorem 3.2.1). Our contribution is not the algebra itself—which is well known in econometrics—but rather its repackaging as an operational diagnostic for the collider/mediator/confounder classification problem in TSCS settings. We begin with the cross-sectional case, which allows the clearest comparison to the classical OVB formula, and then extend the result to ADL models used in TSCS analysis.

4.1 Cross-Section Derivation

Consider the following setup. The true ("short'') model relating the outcome y to the treatment D is:

$$y = \beta_0 + \beta_1 D + e, \qquad \mathbb{E}[e \mid D] = 0 \tag{2}$$

where β_1 is the true causal effect of D on y.

Now suppose a researcher includes an additional variable Z in the regression, estimating the "long'' model:

$$y = \beta_0^\star + \beta_1^\star D + \beta_2^\star Z + e^\star \tag{3}$$

where the star superscript indicates coefficients from the augmented model. Here $\mathbb{E}[e^\star \mid D] \neq 0$ in general when Z is a collider, because conditioning on a common effect opens a spurious path.

To motivate the structural equation for Z, suppose Z is caused by both D and y:

$$Z = \gamma_0 + \gamma_1 D + \gamma_2 y + u, \qquad \mathbb{E}[u \mid D] = 0,\ \mathbb{E}[u \mid y] = 0 \tag{4}$$

The population coefficients from OLS satisfy:

$$\beta_1 = \frac{\mathrm{Cov}(D, y)}{\mathrm{Var}(D)}, \qquad \beta_1^\star = \frac{\mathrm{Cov}(D, y) - \beta_2^\star \,\mathrm{Cov}(D, Z)}{\mathrm{Var}(D)}$$

Subtracting, the change in the estimated treatment effect is:

$$\Delta\beta = \beta_1^\star - \beta_1 = -\beta_2^\star \frac{\mathrm{Cov}(D, Z)}{\mathrm{Var}(D)} \tag{5}$$

To express this in terms of estimable quantities, substitute the true model (2) into the collider equation (4):

$$\begin{aligned} Z &= \gamma_0 + \gamma_1 D + \gamma_2(\beta_0 + \beta_1 D + e) + u \\ &= \underbrace{(\gamma_0 + \gamma_2\beta_0)}_{\phi_0} + \underbrace{(\gamma_1 + \gamma_2\beta_1)}_{\phi_1} D + \underbrace{(\gamma_2 e + u)}_{\varepsilon} \end{aligned} \tag{6}$$

The reduced-form coefficient ϕ_1 is simply the population coefficient from regressing Z on D:

$$\phi_1 = \frac{\mathrm{Cov}(D, Z)}{\mathrm{Var}(D)}$$

Substituting into (5) yields the IVB formula. The following result restates the standard short-versus-long regression decomposition in a form that is directly operational as a diagnostic for covariate inclusion decisions:

Proposition 1 (IVB, Cross-Section)**.** *When an additional variable Z is included in a regression of y on D, the change in the treatment effect estimate is:*

$$\boxed{IVB = \beta_1^\star - \beta_1 = -\beta_2^\star \times \phi_1} \tag{7}$$

where $\beta_2^\star$ is the coefficient on Z in the long regression (3) *and ϕ_1 is the coefficient from regressing Z on D.*

Both $\beta_2^\star$ and ϕ_1 are directly estimable from data: run the long regression to obtain $\hat{\beta}_2^\star$, and regress Z on D to obtain $\hat{\phi}_1$. No knowledge of the structural parameters γ_1 or γ_2 is needed.

Proposition~1 is an algebraic identity—the complement of the standard OVB decomposition, well known in econometrics (e.g., Goldberger, 1991, pp. 197–198). Its value as a contribution lies not in the algebra but in its application. Prior work on collider bias quantification—including path-tracing formulas (Pearl, 2013), structural equation model derivations (Greenland, 2003), and the M-bias sensitivity analysis of Ding and Miratrix (2015)—parameterizes the bias in terms of structural coefficients of the underlying data generating

process, which are not directly observable by the applied researcher. The IVB formula repurposes the known decomposition as a practical diagnostic by expressing the same quantity in terms of regression coefficients the researcher already has in hand ($\hat{\theta}^\star$ from the augmented regression, $\hat{\pi}$ from a simple auxiliary regression). This makes the formula (i)~immediately operational as a diagnostic tool, (ii)~extensible to the ADL models prevalent in TSCS research (Propositions~2–4), and (iii)~a natural mirror of the OVB formula that applied researchers already use routinely.[3]

4.2 OVB versus IVB

The IVB formula has a striking structural parallel with the classical Omitted Variable Bias formula. Table~1 presents the comparison side by side.

Table 1: Comparison of Omitted Variable Bias (OVB) and Included Variable Bias (IVB)

	OVB	**IVB**
Problem	Omitting a confounder	Including a collider
Correct model	Long (includes confounder)	Short (excludes collider)
Wrong model	Short (omits confounder)	Long (includes collider)
Formula	$\delta_1 \times \gamma$	$-\beta_2^\star \times \phi_1$
Components	δ_1: effect of omitted var in long reg	$\beta_2^\star$: effect of collider in long reg
	γ: regression of omitted var on D	ϕ_1: regression of collider on D
Sign of bias	$\text{sign}(\delta_1) \times \text{sign}(\gamma)$	$-\text{sign}(\beta_2^\star) \times \text{sign}(\phi_1)$
Estimable?	Only if confounder is observed	Yes, both components from data

The key difference lies in the direction of the problem. With OVB, the researcher runs a short regression when the long regression is correct, and the bias equals the product of two regression coefficients involving the omitted variable. With IVB, the researcher runs a long regression when the short regression is correct, and the bias is the *negative* product of two analogous coefficients involving the included collider. The negative sign in the IVB formula arises because the mechanisms operate in opposite directions: OVB captures the effect of an omitted common cause channeled through the treatment-outcome relationship, while IVB captures the spurious association created by conditioning on a common effect.

[3]The FWL-based derivation also extends straightforwardly beyond OLS. In two-stage least squares (2SLS) with a single instrument, the second stage is an OLS regression of y on $\hat{D}$ and controls, so the IVB formula applies directly to the second-stage coefficients: $\text{IVB}_{\text{IV}} = -\hat{\theta}^\star_{\text{IV}} \times \hat{\pi}_{\text{IV}}$, where $\hat{\pi}_{\text{IV}}$ is the coefficient from regressing Z on the instrumented treatment $\hat{D}$. For nonlinear models such as logistic regression, the linear probability model (LPM) provides a first-order approximation of the average marginal effect, and the IVB formula applied to the LPM coefficients approximates the collider bias in the AME. We leave formal treatment of these extensions to future work.

A notable practical advantage of IVB is that its components are *always* estimable: the researcher has already included Z in the regression, so $\beta_2^\star$ is available, and ϕ_1 comes from a simple auxiliary regression. By contrast, OVB requires observing the omitted variable—which, by definition, is often unavailable.

Remark (Generality of the IVB formula). *The decomposition $IVB = -\theta^\star \times \pi$ is an algebraic identity of OLS that holds for any variable Z correlated with both D and y, regardless of whether Z is a collider, a confounder, or a mediator. The formula quantifies the arithmetic difference between two nested specifications. Only a DAG or substantive domain expertise determines which specification is causally correct—and therefore whether the difference represents collider bias (exclude Z), omitted variable bias (include Z), or over-control of a mediator (include only Z_{t-1}, not Z_t). This generality makes the IVB formula a unified diagnostic for any covariate inclusion decision, analogous to how the sensitivity analysis of Cinelli and Hazlett (2020) provides a unified diagnostic for omitted variable concerns. The key difference is that IVB addresses inclusion of observed covariates, while Cinelli and Hazlett address omission of unobserved confounders.*

4.3 Extension to Panel Fixed Effects

The cross-sectional result extends immediately to two-way fixed effects (TWFE) models without lagged dependent variables—the workhorse specification in comparative politics. Consider the correct model:

$$y_{it} = \alpha_i + \mu_t + \beta D_{it} + e_{it} \tag{8}$$

and the augmented model including an additional covariate Z_{it}:

$$y_{it} = \alpha_i^\star + \mu_t^\star + \beta^\star D_{it} + \theta^\star Z_{it} + v_{it} \tag{9}$$

By FWL, unit and time fixed effects are absorbed by projecting all variables onto the space spanned by the unit and time dummies. Let $\widetilde{y}_{it}$, $\widetilde{D}_{it}$, and $\widetilde{Z}_{it}$ denote the doubly-demeaned residuals. The argument from Proposition~1 applies to these residuals, yielding:

Corollary 1 (IVB under TWFE)**.** *In a TWFE model without lagged dependent variables, the change in the treatment effect from including Z_{it} is:*

$$IVB = \beta^\star - \beta = -\theta^\star \times \pi \tag{10}$$

where $\theta^\star$ is the coefficient on Z_{it} in the long TWFE regression (9) *and π is the coefficient on D_{it} in the*

auxiliary TWFE regression $Z_{it} = \alpha_i' + \mu_t' + \pi D_{it} + \eta_{it}$.

The corollary has a substantive implication. Because unit fixed effects absorb any time-invariant component of the association between D and Z, the auxiliary coefficient π reflects only the *within-unit* co-movement of treatment and collider. If D and Z are correlated primarily in levels across units (e.g., richer countries have better governance), unit fixed effects remove this correlation and π is small. Similarly, $\theta^\star$ reflects only the within-unit partial association between Z and Y. This is the algebraic basis for the empirical pattern documented in Section~5: IVB tends to be modest in TWFE applications because fixed effects absorb the between-unit collider channels.

4.4 Extension to ADL(1,0) via FWL

We now extend the IVB formula to the dynamic models used in TSCS analysis. Consider an ADL(1,0) specification. The correct ("short'') model is:

$$y_t = \alpha + \beta D_t + \rho y_{t-1} + e_t, \qquad \mathbb{E}[e_t \mid D_t, y_{t-1}] = 0 \tag{11}$$

The augmented ("long'') model includes an additional covariate Z_t:

$$y_t = \alpha^\star + \beta^\star D_t + \rho^\star y_{t-1} + \theta^\star Z_t + v_t \tag{12}$$

Our goal is to express $\beta^\star - \beta$ in terms of estimable quantities. The key tool is the Frisch–Waugh–Lovell (FWL) theorem (Frisch and Waugh, 1933; Lovell, 1963).

Define the set of "legitimate'' controls $W_t = (1, y_{t-1})$ and let $\tilde{y}_t$, $\tilde{D}_t$, and $\tilde{Z}_t$ denote the residuals from projecting each variable on W_t. By FWL, the long regression (12) is equivalent to:

$$\tilde{y}_t = \beta^\star \tilde{D}_t + \theta^\star \tilde{Z}_t + \tilde{v}_t$$

In this bivariate residualized regression, the coefficient on $\tilde{D}_t$ satisfies:

$$\beta^\star = \frac{\mathrm{Cov}(\tilde{D}_t, \tilde{y}_t) - \theta^\star \,\mathrm{Cov}(\tilde{D}_t, \tilde{Z}_t)}{\mathrm{Var}(\tilde{D}_t)}$$

The short model coefficient (also via FWL) is:

$$\beta = \frac{\mathrm{Cov}(\tilde{D}_t, \tilde{y}_t)}{\mathrm{Var}(\tilde{D}_t)}$$

Subtracting:

$$\beta^\star - \beta = -\theta^\star \frac{\mathrm{Cov}(\tilde{D}_t, \tilde{Z}_t)}{\mathrm{Var}(\tilde{D}_t)}$$

Now consider the auxiliary regression:

$$Z_t = a + \pi D_t + \lambda y_{t-1} + \eta_t \tag{13}$$

By the algebra of partial regression, $\pi = \mathrm{Cov}(\tilde{D}_t, \tilde{Z}_t)/\mathrm{Var}(\tilde{D}_t)$. Substituting, the same decomposition applies in the dynamic setting:

Proposition 2 (IVB, ADL(1,0))**.** *In an ADL(1,0) model, when an additional covariate Z_t is included, the change in the treatment effect estimate is:*

$$\boxed{IVB(\beta) = \beta^\star - \beta = -\theta^\star \times \pi} \tag{14}$$

where $\theta^\star$ is the coefficient on Z_t in the long regression (12) *and π is the coefficient on D_t in the auxiliary regression* (13)*.*

The formula has the same structural form as the cross-sectional case. The only difference is that all quantities are now *partial* effects, conditioned on the legitimate controls $(1, y_{t-1})$. The dynamics enter through the residualization, but the final formula remains a simple product.

4.5 Generalization to ADL(p,q)

The result extends naturally to general ADL(p,q) specifications. Consider the correct model:

$$y_t = \alpha + \sum_{i=1}^{p} \rho_i y_{t-i} + \sum_{j=0}^{q} \beta_j D_{t-j} + e_t$$

and the augmented model including Z_t:

$$y_t = \alpha^\star + \sum_{i=1}^{p} \rho_i^\star y_{t-i} + \sum_{j=0}^{q} \beta_j^\star D_{t-j} + \theta^\star Z_t + v_t$$

Let W_t denote the vector of all legitimate controls (constant, $y_{t-1}, \ldots, y_{t-p}$, and the relevant lags of D other than D_{t-j}). The FWL-based decomposition extends immediately:

Proposition 3 (IVB, ADL(p,q))**.** *For any coefficient β_j in an ADL(p,q) model, the change from including an additional covariate Z_t is:*

$$\boxed{IVB(\beta_j) = \beta_j^\star - \beta_j = -\theta^\star \times \pi_j} \tag{15}$$

where π_j is the coefficient on D_{t-j} in the auxiliary regression $Z_t = a + \pi_j D_{t-j} + W_t'\delta + \eta_t$.

The practical value of this decomposition lies in its uniformity across specifications. Regardless of the number of lags in the outcome or the treatment, the IVB for any coefficient β_j is always the negative product of two estimable quantities: the collider coefficient $\theta^\star$ in the long model, and the partial association π_j between the treatment at lag j and the collider, controlling for all other legitimate regressors.

An important practical observation is that while the *form* of the IVB is simple, the *magnitude* can vary substantially depending on the dynamics. When the treatment D_t is highly predictable from lags, the residualized treatment $\tilde{D}_t$ has little variation, and π can become unstable. When Z_t is a post-treatment variable that responds to y or to correlated shocks, both $\theta^\star \neq 0$ and $\pi \neq 0$ tend to arise mechanically. Two additional results are developed in the appendix: the interaction between IVB and Nickell bias in short panels (Appendix~A.3), and the fact that replacing the treatment with a longer lag does not mechanically eliminate the IVB when the treatment is autocorrelated (Appendix~A.4).

4.6 A Practical Recipe

We summarize the IVB formula as a simple diagnostic procedure for the applied researcher:

Step 0. Establish causal status. Using a DAG or substantive domain expertise, establish that Z is plausibly a collider—a variable caused by both D and Y (or by D and an unobserved common cause of Y). If Z is instead a confounder, the IVB formula quantifies omitted variable bias in the short model, and the *long* model is correctly specified. The steps below are informative as a collider diagnostic only if Z's collider status has been established through causal reasoning, not statistical testing. In Section 6, we illustrate how to combine literature review with DAG reasoning to assess collider plausibility before computing the IVB.

1. **Run the long regression.** Estimate the model that includes the suspected collider Z:

$$y = \ldots + \theta^\star Z + \text{other controls} + v$$

Record $\hat{\theta}^\star$.

2. **Run the auxiliary regression.** Regress the suspected collider on the treatment and all legitimate controls:

$$Z = \ldots + \pi D + \text{same controls as in step 1, minus } Z + \eta$$

Record $\hat{\pi}$.

3. **Compute the bias.** The estimated IVB is:

$$\widehat{\text{IVB}} = -\hat{\theta}^\star \times \hat{\pi}$$

The uncertainty of $\widehat{\text{IVB}}$ can be quantified via the delta method or cluster bootstrap; see Appendix A.7 for details and an upper bound on $\text{SE}(\widehat{\text{IVB}})$ that requires only standard regression output.

If $|\widehat{\text{IVB}}|$ is large relative to the estimated treatment effect, the researcher has strong reason to exclude the collider from the specification. The sign of the IVB indicates the *direction* of the bias: whether including the collider attenuates or amplifies the treatment effect.

4.7 Interpretation Caveats

An important caveat concerns the interpretation of the IVB formula. The decomposition $\text{IVB} = -\theta^\star \times \pi$ is an **algebraic identity**: it holds whenever the short and long regressions are nested, regardless of the causal role of Z. The formula quantifies the *arithmetic* difference between two nested specifications; it does not, by itself, determine which specification is causally correct. The product $\theta^\star \times \pi$ will be non-zero whenever Z is correlated with both D and y conditional on other controls—a condition satisfied not only when Z is a collider, but also when Z is a confounder or a mediator.

The three cases lead to fundamentally different conclusions:

- Z **is a collider.** Then the short regression is correctly specified, the IVB formula gives the bias introduced by including Z, and the researcher should exclude Z.
- Z **is a confounder.** Then the long regression is correctly specified, and the same algebraic decomposition gives the omitted variable bias in the short regression. In this case, excluding Z *introduces* bias rather than removing it. The formula is still informative—it tells the researcher how sensitive the treatment estimate is to including Z—but its sign is reversed: a large IVB is evidence that Z is an important confounder, not that it is a harmful collider.

- **Z is a contemporaneous mediator.** Then Z_t lies on the causal path $D_t \to Z_t \to Y_t$, and the long regression recovers the direct effect while the short regression recovers the total effect. The IVB formula captures the difference between the two—i.e., the indirect effect mediated by Z_t. In TSCS settings where the estimand is the contemporaneous total effect (CET), the researcher should avoid conditioning on Z_t and instead condition on the lagged value Z_{t-1}, which captures confounding without blocking the contemporaneous causal channel (Section 3.4).

Only the DAG determines which case applies. The IVB formula complements DAGs but cannot substitute for them: the graph tells the researcher *whether* Z is a collider, and only then does the formula quantify how much damage including it causes. We illustrate this interplay in Section~6, where we combine literature review with DAG reasoning to assess collider plausibility before computing the IVB. The Rogowski et al. application in particular shows that a large IVB does not automatically imply collider bias—it may instead reflect appropriate deconfounding.

Sufficient conditions for classifying Z as a collider include: (i)~credible evidence from separate studies that D causally affects Z and Y causally affects Z; (ii)~absence of evidence that Z causally affects either D or Y; and (iii)~the causal effect of D on Z and Y on Z operating through mechanisms that are not mediated by the other. When these conditions are not met—for instance, when Z is simultaneously a confounder and a collider—the IVB formula still quantifies the arithmetic difference between specifications, but its causal interpretation requires additional assumptions (see Ding and Miratrix 2015 for the "Butterfly-Structure'' case).

The empirically prevalent case in which Z is simultaneously a collider and a confounder deserves particular attention. Ding and Miratrix (2015) analyzed one such configuration—the "Butterfly-Structure''—in cross-sectional LSEMs, finding that conditioning on Z reduces net bias in approximately 75% of the feasible parameter space. However, this result is specific to a single structural configuration with particular symmetry assumptions. In TSCS settings with fixed effects, the landscape is richer: FE absorbs between-unit confounding while leaving within-unit collider channels intact, potentially shifting the trade-off in ways that depend on the panel structure. A systematic FWL-based derivation of when collider bias dominates confounding bias under TWFE—yielding analytical results rather than relying solely on simulation—is a promising direction for future work that would extend both our framework and that of Ding and Miratrix (2015) to the panel setting.

5 Benchmarking Dynamic Specifications for the CET

The IVB formula tells a researcher *how much* the treatment estimate changes when a covariate is included. The DAG tells the researcher *what* that change means. Together, they raise a practical question: which specification should a TSCS researcher adopt when the estimand is the contemporaneous treatment effect (CET)?

We use Monte Carlo simulations spanning 238 DGP configurations and 500 replications each to evaluate specifications across three empirically relevant scenarios: collider and dual-role controls, contemporaneous mediators, and robustness to nonlinearity and feedback. The evidence converges on a single recommendation: for the CET in TSCS settings, ADL + FE with lagged state variables provides a simple and defensible benchmark.

5.1 Collider and Dual-Role Controls

The most common scenario in TSCS applications is a covariate Z that plays a dual role: Z_{t-1} is a confounder (it affects both D_t and Y_t through persistence), while Z_t is a collider ($D_t \to Z_t \leftarrow Y_t$). The researcher faces a dilemma: excluding Z entirely leaves confounding from Z_{t-1} unaddressed, while including the contemporaneous Z_t introduces collider bias.

Two features of TWFE already attenuate the collider channel. First, as shown in Corollary~1, the IVB under TWFE depends only on the within-unit components of $\theta^\star$ and π. If the association between D and Z is primarily cross-sectional (e.g., richer countries have better governance in levels), unit fixed effects absorb this association and π is small. An analogous argument applies to $\theta^\star$. Second, in studies with binary, slow-moving treatments (democratization, peacekeeping deployment, policy adoption), few "switchers'' change treatment status, inflating SE($\hat{\beta}$) while the population IVB remains roughly constant. In Monte Carlo simulations ($N = 200$, $T = 30$, $\beta = 1$, 500 replications), reducing the fraction of switchers from 70% to 10% increases the standard error by a factor of 2–3, reducing |IVB/SE| proportionally.[4]

But TWFE alone does not resolve the dual-role case. To quantify the gain from dynamic specifications, we simulate a DGP with exogenous binary treatment (staggered adoption) and a Z variable that is both a confounder ($Z_{t-1} \to Y_t$) and a collider ($D_t \to Z_t$, $Y_t \to Z_t$) with persistence ($Z_{t-1} \to Z_t$). We estimate nine specifications ranging from TWFE without controls to ADL models with all available lags (Y_{t-1}, D_{t-1}, Z_{t-1}). Across 48 scenarios ($N = 100$, $T = 30$, 500 replications), TWFE without Z exhibits bias of 77–169%

[4] An important caveat: the fact that |IVB/SE| is small does not mean the bias is harmless. The population IVB is constant regardless of the number of switchers. In a study with few switchers and a marginally significant result, the IVB could still be large enough relative to $\hat{\beta}$ to flip the sign or significance of the treatment effect. The researcher should therefore consider both $|\text{IVB}/\hat{\beta}|$ (the bias as a share of the estimated effect) and |IVB/SE| (the bias relative to sampling uncertainty).

of β—driven by omitted variable bias from the confounding channel of Z. The ADL specification including Y_{t-1}, D_{t-1}, and Z_{t-1} reduces bias to less than 3% of β in every scenario.

The key insight is that the ADL specification resolves the dual-role dilemma by exploiting the temporal structure: Z_{t-1} captures the confounding channel (because Z_{t-1} causally affects both D_t and Y_t through persistence), while Y_{t-1} in the regression blocks the collider path $D_t \to Z_t \leftarrow Y_t$ that would otherwise transmit bias through Z's persistence. The result is consistent with the identification strategy in Imai and Kim (2021), where the ADL model achieves sequential ignorability under weaker assumptions than TWFE.

5.2 Over-Control with Contemporaneous Mediators

When Z_t is a contemporaneous mediator ($D_t \to Z_t \to Y_t$) rather than a collider, a different problem arises. Including Z_t does not open a spurious path; instead, it blocks the indirect causal channel, so the regression recovers only the direct effect of D_t on Y_t. If the researcher's estimand is the total contemporaneous effect (CET), this constitutes over-control bias: the estimate is attenuated by the indirect effect mediated by Z_t.

To quantify this, we simulate a DGP with a pure contemporaneous mediator ($D_t \to Z_t \to Y_t$, $\beta = 1$, $\theta = 0.5$, $\delta_D = 0.4$), so the true total effect is $1.2 = 1.0 + 0.5 \times 0.4$. We compare three ADL specifications: `ADL total'' (without $Z$),`ADL bad'' (with Z_t contemporaneous), and "ADL safe'' (with Z_{t-1} only). Across nonlinearity types and collider persistence levels ($N = 200$, $T = 30$, 500 replications):

- ADL total recovers the CET: $\hat{\beta} \approx 1.20$ (bias $< 0.1\%$).
- ADL bad recovers the direct effect: $\hat{\beta} \approx 1.00$, with over-control of ≈ 0.20 (the indirect effect).
- ADL safe preserves the CET: $\hat{\beta} \approx 1.20$ (bias $< 0.2\%$), because Z_{t-1} is pre-treatment for the contemporaneous effect and does not block $D_t \to Z_t \to Y_t$.

The IVB formula captures this over-control exactly: $\text{IVB} = -\hat{\theta}^\star \times \hat{\pi} \approx -0.20$, coinciding with the indirect effect under linearity.

A richer case arises when Z is simultaneously a contemporaneous mediator and a confounder through lags ($Z_{t-1} \to D_t$, $Z_{t-1} \to Y_t$). In this configuration, the researcher *needs* Z to satisfy conditional parallel trends, but including Z_t would generate over-control. The ADL with Z_{t-1} resolves this tension: it removes confounding from Z_{t-1} (satisfying conditional parallel trends) without blocking the contemporaneous causal channel $D_t \to Z_t \to Y_t$. In simulations with confounding parameters $\gamma_D = 0.15$, $\gamma_Y = 0.2$: excluding Z entirely yields bias of 6–11% of β (OVB from the confounding channel); including Z_t reduces bias to 2% but attenuates the estimate (over-control of the indirect effect); including only Z_{t-1} achieves both goals, with

bias of approximately 3% and no over-control of the contemporaneous channel.

This result highlights a structural advantage of the TSCS setting over the canonical two-period DID. In DID with two periods, the researcher has only the contemporaneous Z_{t^*}: including it satisfies conditional parallel trends but generates over-control; excluding it avoids over-control but violates conditional parallel trends. This is the dilemma motivating doubly robust and imputation-based methods in the DID literature (Caetano et al., 2022). In TSCS, the temporal structure separates the two roles of Z: the lagged value absorbs confounding, and the contemporaneous value is left out of the regression, preserving the total effect. Appendix~A.6 formalizes this argument by re-deriving the IVB formula in the potential outcomes framework and connecting it to the bias decomposition of Caetano and Callaway (2024).

5.3 Robustness

The preceding results are derived under linearity. Two departures from linearity could threaten the ADL benchmark: nonlinearity in the collider or mediator equation, and feedback from outcomes to treatment.

Nonlinearity. We simulate three classes of nonlinearity in the Z equation: (i)~concave transformations ($\log(1 + |D|)$, soft-clamped polynomial), (ii)~polynomial terms (D^2), and (iii)~interaction terms ($D \times H$ with H exogenous). For bounded nonlinearities—concave or saturating transformations that are realistic for most political science applications—the ADL model with all lags remains robust: bias stays below 3% of β for the collider case and over-control remains within 5% of the linear baseline for the mediator case. For unbounded nonlinearities (e.g., high-degree polynomials with large coefficients), the DGP can become explosive. However, such configurations are empirically implausible: they require Z to respond without bound to marginal increases in D, which contradicts the ceiling and floor effects that characterize most real-world variables.

Feedback. When the outcome causally affects future treatment ($Y_{t-1} \to D_t$), strict exogeneity is violated (Imai and Kim, 2021). We simulate this by adding a feedback parameter $\phi \in \{0, 0.05, 0.1, 0.15\}$ to the treatment equation. TWFE bias grows monotonically with ϕ: from 43% of β at $\phi = 0$ to 80% at $\phi = 0.15$. The ADL specification with all lags remains robust: $|\text{bias}| \leq 1\%$ of β across all stable scenarios, because Y_{t-1} in the regression absorbs the feedback channel. For $\phi \geq 0.18$, the joint dynamic system becomes explosive, marking a boundary beyond which no OLS-based estimator is adequate.

5.4 Practical Benchmark for the CET

The simulation evidence converges on a practical rule for TSCS researchers interested in the contemporaneous treatment effect:

> *When the estimand is the CET, use ADL + FE with lagged state variables as the benchmark specification. Include Z_{t-1} when the lagged covariate is substantively appropriate (confounder or dual role). Avoid conditioning on contemporaneous Z_t when Z_t may be post-treatment—either a collider or a mediator in the period of interest.*

This recommendation is consistent with and reinforces the qualitative guidance of Blackwell and Glynn (2018), who showed that the CET (β_1) is consistent in the ADL under linearity (p. 1073), and of Imai and Kim (2021), who recommend ADL + FE as the parametric solution for dynamic panel identification (Table 1, rows 2–4). Our contribution is to provide (a) a quantitative diagnostic formula for the change caused by specification choices, and (b) systematic Monte Carlo evidence across 238 DGP configurations—spanning collider, dual-role, and mediator cases with nonlinearities and feedback—that the ADL benchmark works uniformly for these cases. Specifically, the rule is supported by three results. First, for collider and dual-role controls, ADL + FE reduces bias to less than 3% of β across all 48 dual-role scenarios, compared to 77–169% for TWFE without controls. Second, for contemporaneous mediators, conditioning on Z_{t-1} instead of Z_t preserves the total effect while still absorbing confounding through lags—a timing insight that, to our knowledge, has not been systematically demonstrated in prior work. Third, the results are robust to bounded nonlinearities and moderate outcome-to-treatment feedback.

The rule has explicit limits. It applies to the contemporaneous effect of D_t on Y_t; for lagged, cumulative, or long-run effects, Z_{t-1} may itself be post-treatment and the timing argument does not apply. It requires that the DAG be at least approximately known—the researcher must classify each covariate as collider, confounder, mediator, or dual role before applying the benchmark. And it does not claim to replace more flexible estimators (AIPW, doubly robust) in settings where the ADL's linearity assumption is strongly violated. Extending the timing logic developed here to synthetic-control-style and interactive fixed-effects estimators is an important agenda for future work.

6 Empirical Applications

We now illustrate how the IVB formula and the DAG-based classification work together in practice. We apply the diagnostic to six published studies spanning several substantive domains: democratization and public opinion (Claassen, 2020), democracy and ethnic inequality (Leipziger, 2024), UN peacekeeping and

democratization (Blair, Di Salvatore and Smidt, 2022), fiscal capacity in Africa (Albers, Jerven and Suesse, 2023), postal infrastructure and growth (Rogowski et al., 2022), and sovereign bond denomination (Ballard-Rosa, Mosley and Wellhausen, 2021). The studies were selected to cover a range of treatments (binary and continuous), outcome types, and subfields within comparative politics and international political economy, with the requirement that replication data and a TWFE or panel FE specification be publicly available. For each study, we replicate the main specification and compute the IVB decomposition for every control variable. The goal is not merely to catalog IVB magnitudes, but to show how a researcher can use the formula to evaluate whether a specific control improves or worsens the specification—and by how much.

Consistent with the diagnostic recipe in Section~4.6, we begin with Step~0: for each study, we classify every control variable as a collider, confounder, mediator, or predetermined/neutral variable, drawing on the published literature to assess whether credible evidence exists for both the $D \to Z$ and $Y \to Z$ channels. A control is classified as a collider candidate only when separate studies provide credible evidence that the treatment causes the control *and* the outcome causes the control. The detailed classification tables, with supporting references, are reported in Appendix~A.8. The full mechanical IVB decomposition for all 55 study-control combinations—regardless of collider plausibility—is also available in Appendix~A.8.

Table~2 reports the IVB only for controls classified as collider candidates. We use two complementary metrics. The primary metric is $|\text{IVB}/\text{SE}(\hat{\beta})|$, which measures the bias in standard-error units. This metric is always well-defined (the standard error is bounded away from zero for any valid estimate) and has a natural benchmark: $|\text{IVB}/\text{SE}| > 1$ means the bias exceeds the sampling uncertainty of the treatment effect. As a secondary metric, we report $\text{IVB}/\hat{\beta}$ (in percent) when the treatment effect is statistically significant ($|t| \geq 1.96$), which gives the bias as a share of the estimated effect. When the treatment effect is not significant, $\text{IVB}/\hat{\beta}$ is reported as "—'' because dividing by a quantity indistinguishable from zero produces uninformative ratios.

Three patterns emerge from the table. First, among the 14 collider candidates identified across six studies, only one produces an IVB exceeding one standard error of the treatment effect: log GDP per capita in Rogowski et al. (2022), with $|\text{IVB}/\text{SE}| = 2.11$, corresponding to 58% of $\hat{\beta}$. This is the only case in which the covariate inclusion decision is quantitatively consequential. Second, the median $|\text{IVB}/\text{SE}|$ across collider candidates (excluding Albers, Jerven and Suesse 2023, where standard errors are unavailable) is approximately 0.13—the typical IVB is about one-eighth of a standard error, well within sampling noise. This is consistent with the simulation evidence in Section~5: fixed effects absorb the between-unit collider channels, and the residual within-unit IVB is modest. Third, GDP per capita and related economic variables dominate the collider candidates across studies, consistent with their dual role as common effects of political

Table 2: IVB for theoretically motivated collider candidates across six studies. Controls are classified as collider candidates based on DAG reasoning (see Appendix A.8). IVB/SE measures the bias in standard-error units; values exceeding 1 indicate that the bias surpasses sampling uncertainty. IVB/$\hat{\beta}$ is reported only when the treatment effect is statistically significant ($|t| \geq 1.96$); "—" indicates that $\hat{\beta} \approx 0$ renders the ratio uninformative. For Albers et al., standard errors are unavailable from the published replication, so IVB/SE is not reported. The full mechanical decomposition for all 55 study-control combinations is in Appendix A.8.

Study	Collider candidate	IVB	|IVB/SE|	IVB/$\hat{\beta}$ (%)
Claassen (2020, FE)	Log GDP p.c.	-0.0165	-0.08	—
Leipziger (2024, SEI)	Log GDP p.c.	0.0056	0.48	-15.9%
Leipziger (2024, ext.)	Civil war	0.0002	0.02	-0.7%
Leipziger (2024, ext.)	GDP growth	-0.0000	-0.00	0.0%
Blair et al. (2022)	Foreign Aid	0.0029	0.07	2.4%
Blair et al. (2022)	GDP per capita	0.0055	0.14	4.7%
Blair et al. (2022)	Refugees/IDPs	0.0050	0.13	4.2%
Albers et al. (2023)	Hyperinflation	0.0288	—	—
Albers et al. (2023)	GDP growth (YoY)	0.0163	—	—
Albers et al. (2023)	Liberal democracy	-0.0131	—	—
Rogowski et al. (2022)	Log GDP p.c.	0.0115	2.11	58.0%
Ballard-Rosa et al. (2021)	IMF program	-0.0086	-0.19	-7.9%
Ballard-Rosa et al. (2021)	Inflation crisis	-0.0125	-0.28	-11.4%
Ballard-Rosa et al. (2021)	Sov. debt crisis	-0.0038	-0.09	-3.5%

treatments and socioeconomic outcomes—and illustrating why the DAG-based classification step is essential before interpreting the formula.

The IVB/SE metric avoids the pitfalls of percentage-of-$\hat{\beta}$ ratios when the treatment effect is near zero. For instance, in Claassen's FE specification, the treatment effect is not statistically significant ($\hat{\beta} = -0.016$, SE $= 0.216$), making IVB/$\hat{\beta}$ uninformative. The IVB/SE metric reveals that the bias is just 8% of a standard error ($|\text{IVB/SE}| = 0.08$)—completely negligible. When the treatment effect *is* significant, IVB/$\hat{\beta}$ provides useful complementary information about substantive magnitude: in Rogowski et al. (2022), the IVB of 58% of $\hat{\beta}$ indicates that more than half the estimated effect could reflect collider bias (though it could equally reflect appropriate deconfounding—see Section~6.2).

A caveat on cross-study comparisons: standard errors vary substantially across studies (from 0.005 in Rogowski et al. to 0.216 in Claassen's FE specification), reflecting differences in sample size, clustering structure, and treatment variation rather than differences in bias magnitude. The |IVB/SE| benchmark is most informative *within* a given study—as a measure of whether collider bias exceeds that study's own sampling uncertainty—and should be interpreted with caution when comparing across studies with very different precision levels.

These aggregate results establish the empirical baseline. We now examine two studies in detail to illustrate

how the IVB formula interacts with DAG reasoning in practice. In Leipziger (2024), the IVB is modest and the collider interpretation is well supported—the formula confirms that the specification choice is quantitatively inconsequential. In Rogowski et al. (2022), the IVB is large but causally ambiguous—the formula reveals that the specification choice matters substantially, but only the DAG can determine which specification is correct. Together, the two cases demonstrate both the power and the limits of the diagnostic. Full replication details for the remaining four studies are available in the Online Appendix.

6.1 Application 1: Democracy and Ethnic Inequality

Leipziger (2024) estimates the effect of democratic transitions on socioeconomic ethnic inequality using TWFE:

$$\text{Ethnic Inequality}_{it} = \beta\,\text{Democracy}_{i,t-1} + \theta\,\text{GDP p.c.}_{i,t-1} + \delta_i + \gamma_t + \varepsilon_{it}$$

where ethnic inequality is measured by three indices (V-Dem public services, nightlight-based income from Alesina et al., and education from Omoeva et al.), Democracy is a binary indicator based on the Lexical Index (≥ 5), and GDP per capita is the sole time-varying control in the baseline specification. Leipziger deliberately excludes additional controls from the baseline, citing "the risk of posttreatment bias'' (p. 1347)—a concern consistent with the collider bias framework developed in this paper.

The replicated treatment coefficient is $\hat{\beta} = -0.035$ (SE $= 0.012$, $N = 15,191$), indicating that democratization reduces ethnic inequality in public services.

6.1.1 Collider Assessment

Is GDP per capita a collider in this setting? The question requires consulting two literatures.

First, does democracy (the treatment) cause GDP per capita? The evidence is strong. Acemoglu et al. (2019) use dynamic panel methods, system GMM, and regional democratization waves as instruments to estimate that democratization increases GDP per capita by approximately 20% over 25 years, operating through investment, education, and reduced social conflict. Papaioannou and Siourounis (2008) find approximately 1 percentage point of additional annual growth following democratization.

Second, does ethnic inequality (the outcome) cause GDP per capita? The evidence is also strong. Easterly and Levine (1997) showed that ethnic diversity explains poor growth performance in Africa through political instability and underinvestment in public goods. Grundler and Link (2024) use instrumental variables based on the artificiality of subnational borders in Africa to estimate a *causal* effect: moving from an equal distribution to complete concentration reduces GDP per capita by 12–15%.

Both channels—$D \to Z$ and $Y \to Z$—are supported by credible causal evidence, making GDP per capita a plausible collider. This is an instance of foreign collider bias: the researcher studying ethnic inequality must consult the growth economics literature to discover that GDP per capita is not merely a confounder but also a collider.

6.1.2 IVB Decomposition

Table 3: IVB decomposition for Leipziger (2024): Democracy → Ethnic Inequality (SEI), with GDP per capita as candidate collider.

	Estimate
$\hat{\beta}_{\text{short}}$ (without GDP p.c.)	-0.0407
$\hat{\beta}_{\text{long}}$ (with GDP p.c.)	-0.0352
$\hat{\theta}^*$ (GDP p.c. in long model)	-0.0497
$\hat{\pi}$ (Democracy in auxiliary reg.)	0.1123
IVB $= -\hat{\theta}^* \times \hat{\pi}$	0.0056
|IVB/SE|	0.48
$\|\text{IVB}\|/|\hat{\beta}_{\text{long}}|$	15.9%

GDP per capita produces an IVB of approximately 0.0056, corresponding to |IVB/SE| = 0.48—about half a standard error—and approximately 16% of the treatment effect. The direction is consistent with the literature: $\hat{\theta}^\star < 0$ (richer countries have less ethnic inequality) and $\hat{\pi} > 0$ (democracy increases income), so IVB $= -\hat{\theta}^\star \hat{\pi} > 0$, pushing the treatment coefficient toward zero. In other words, controlling for GDP per capita *attenuates* the estimated effect of democratization on ethnic inequality. The short model (without GDP per capita) yields a somewhat larger effect.

In the extended specification that includes additional controls, civil war is a second collider candidate: democracy reduces civil war risk (Cederman, Gleditsch and Buhaug, 2013) and ethnic inequality increases it (Cederman, Gleditsch and Buhaug, 2013), satisfying both the $D \to Z$ and $Y \to Z$ channels. However, its IVB is negligible (|IVB/SE| = 0.02).

This application illustrates the IVB diagnostic in a clean case. First, even a single control retained for valid confounding-adjustment reasons can introduce a measurable IVB when it is also a collider. Leipziger's careful exclusion of other controls to avoid post-treatment bias was well motivated—but the one control that was retained (GDP per capita) is itself a candidate collider. Second, the IVB of approximately 16%

is substantively notable but does not threaten the main finding: at half a standard error, it falls well within sampling uncertainty. This is consistent with the benchmark logic of Section~5: TWFE absorbs the between-unit component of the democracy–GDP association, leaving only the modest within-unit co-movement to drive the IVB. Note that Leipziger uses lagged controls (GDP p.c.$_{i,t-1}$), which is the safe timing choice identified in Section~3.4: the lagged value captures confounding without opening a contemporaneous collider or mediator path.

6.2 Application 2: Postal Infrastructure and Economic Growth

Rogowski et al. (2022) estimate the effect of public postal infrastructure on economic development using a TWFE country panel model. The main specification is:

$$\text{GDP growth}_{i,t+1} = \beta\,\text{Post Offices}_{i,t-1} + \mathbf{Z}_{it}'\gamma + \delta_i + \mu_t + \varepsilon_{it}$$

where GDP growth is the 5-year growth rate (one period forward), Post Offices is the log cumulative stock (5-year lag), and $\mathbf{Z}_{it}$ includes four controls: log GDP per capita (5-year lag), log population, urbanization rate, and Polity2 democracy score.

The replicated treatment coefficient is $\hat{\beta} = 0.0198$ (SE $= 0.0054$, $N = 1,540$). Post offices have a statistically significant positive effect on subsequent GDP growth.

6.2.1 Collider Assessment

The key collider candidate is log GDP per capita. The paper's core argument is that post offices facilitate economic development ($D \to Z$)—indeed, this is precisely the mechanism being tested. At the same time, GDP growth (the outcome) mechanically determines future GDP per capita levels ($Y \to Z$, through the accounting identity that growth cumulates into levels). This creates a collider structure: both D and Y cause Z.

However, GDP per capita is *simultaneously* a strong confounder in growth regressions, through conditional convergence: richer countries grow more slowly, holding other factors constant. This is a well-established empirical regularity in the growth literature (Acemoglu et al., 2019). Whether controlling for GDP per capita introduces collider bias (through the $D \to Z$ and $Y \to Z$ channels) or removes confounding bias (through the convergence channel) depends on the relative strength of these effects—a question the IVB formula alone cannot answer.

6.2.2 IVB Decomposition

Table 4: IVB decomposition for Rogowski et al. (2022): Post Offices → GDP Growth.

Control as z	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{long}}$	$\hat{\theta}^*$	$\hat{\pi}$	IVB	|IVB/SE|	IVB/$\hat{\beta}$ (%)
Log GDP p.c.	0.0083	0.0198	-4.0130	0.0029	0.0115	2.11	58.0
Log population	0.0206	0.0198	-1.9465	-0.0004	-0.0008	0.15	-4.0
Urbanization	0.0192	0.0198	-3.1421	0.0002	0.0006	0.11	3.0
Polity2	0.0191	0.0198	0.0341	-0.0221	0.0008	0.14	3.8

Log GDP per capita dominates the IVB decomposition, with |IVB/SE| = 2.11—the only case across all six studies where the collider bias exceeds one standard error of the treatment effect. This corresponds to approximately 58% of $\hat{\beta}_{\text{long}}$. Including GDP per capita *inflates* the treatment effect from 0.0083 (without GDP p.c.) to 0.0198 (with GDP p.c.). The mechanism is clear: $\hat{\theta}^\star < 0$ (convergence—higher GDP per capita predicts lower subsequent growth) and $\hat{\pi} > 0$ (more post offices predict higher GDP per capita), so IVB $= -\hat{\theta}^\star \hat{\pi} > 0$. The other controls produce IVBs well below one standard error.

6.2.3 The Limits of the IVB Formula

This application illustrates the crucial caveat from Section~4.7: **a large IVB does not automatically imply collider bias**. The 58% IVB for GDP per capita could reflect either of two causal structures:

- If GDP per capita is primarily a **collider** (because post offices raise GDP through the paper's core mechanism, and growth mechanically cumulates into GDP levels), then including it biases the treatment effect upward, and the short model's estimate of 0.0083 is closer to the truth.
- If GDP per capita is primarily a **confounder** (because richer countries both build more post offices and grow more slowly due to convergence), then excluding it biases the treatment effect downward, and the long model's estimate of 0.0198 is closer to the truth.

In reality, GDP per capita is likely *both* a confounder and a collider, and the net effect depends on the

relative magnitudes of these channels. The IVB formula cannot resolve this ambiguity—it can only quantify the arithmetic difference between the two specifications. The researcher must use domain knowledge and causal reasoning to determine which specification is more appropriate.

This case underscores that the IVB formula is a *diagnostic* tool, not a decision rule. It tells the researcher exactly how much the treatment estimate changes when a covariate is included. Whether that change represents the introduction of bias (collider), the removal of bias (confounder), or over-control of a mediator requires a DAG. The Rogowski et al. case is precisely the dual-role scenario discussed in Section~5.1: the researcher needs domain knowledge to determine the net causal role before deciding whether to include, exclude, or lag the covariate.

7 Conclusion

Applied researchers using TSCS data routinely face a practical question: should a time-varying covariate be included as a control? The answer depends on the covariate's causal role—confounder, collider, or mediator—and on its timing relative to the treatment. This paper provides two tools for navigating that decision.

The first is the IVB formula, $\text{IVB} = -\theta^\star \times \pi$, which quantifies the arithmetic difference between specifications that include and exclude a covariate. Both components are directly estimable from data: $\hat{\theta}^\star$ from the regression that includes the covariate, and $\hat{\pi}$ from a simple auxiliary regression. The formula is an algebraic identity of OLS that holds in cross-sectional, TWFE, and ADL models, for any treatment type. Combined with a DAG, it tells the researcher whether the difference represents collider bias (exclude the covariate), omitted variable bias (include it), or over-control of a mediator (include only its lagged value).

The second is a practical benchmark for the contemporaneous treatment effect (CET): use ADL + FE with lagged state variables as the default specification. Include Z_{t-1} when the lagged covariate is substantively appropriate (confounder or dual role). Avoid conditioning on contemporaneous Z_t when it may be post-treatment—either a collider or a mediator in the period of interest. This recommendation reinforces the qualitative guidance of Blackwell and Glynn (2018) and Imai and Kim (2021); our contribution is the quantitative diagnostic formula and systematic Monte Carlo validation across 238 DGP configurations, which confirm that this benchmark reduces collider bias to less than 3% of β, preserves the total contemporaneous effect when covariates are lagged rather than contemporaneous, and is robust to bounded nonlinearities and moderate outcome-to-treatment feedback. Across six published studies spanning democratization, peacekeeping, fiscal capacity, infrastructure, and sovereign debt, the median |IVB/SE| among collider candidates

is approximately 0.13—well within sampling noise.

The temporal structure of TSCS data provides a structural advantage for this problem. In the canonical two-period DID, a covariate that is both a confounder and a contemporaneous mediator presents an inescapable dilemma: including it satisfies conditional parallel trends but blocks the indirect causal channel; excluding it avoids over-control but leaves confounding unaddressed. In TSCS, the lagged value Z_{t-1} separates the two roles: it absorbs confounding without blocking the contemporaneous path $D_t \to Z_t \to Y_t$. This is the timing logic that the ADL benchmark exploits, and it connects to the active research on bad controls in the DID literature (Caetano et al., 2022)—showing that in TSCS settings focused on the CET, a simpler solution already exists.

Several limitations bound these results. The benchmark applies to the contemporaneous effect of D_t on Y_t; for lagged, cumulative, or long-run effects, Z_{t-1} may itself be post-treatment and the timing argument does not apply. The IVB formula is derived for linear models; while bounded nonlinearities preserve its practical relevance, the Frisch–Waugh–Lovell theorem does not hold for nonlinear link functions. The formula quantifies the *arithmetic* difference between nested specifications but cannot determine which specification is causally correct—only a DAG or substantive domain expertise can do that. And we do not develop a formal sensitivity analysis framework for IVB, which would allow researchers to assess how sensitive their results are to potential collider bias from unobserved variables.

These limitations point to three directions for future work. First, just as Cinelli and Hazlett (2020) developed sensitivity tools that extend OVB to unobserved confounders, a parallel development for IVB would allow researchers to ask: "How strong would the collider relationship need to be to change my conclusions?'' Second, extending the timing logic developed here to synthetic-control-style and interactive fixed-effects estimators is an important agenda—the ADL benchmark covers linear parametric models, but the growing use of SCM, SDiD, and factor models in comparative politics calls for analogous results in those frameworks. Third, an FWL-based derivation of when collider bias dominates confounding bias under TWFE—yielding closed-form conditions for the dual-role case—would extend both our framework and that of Ding and Miratrix (2015) to the panel setting.

We provide the `compute_ivb_multi()` function as supplementary material, enabling researchers to compute IVB decompositions for any TWFE or ADL specification with minimal effort. The combination of DAGs for classifying covariates, the IVB formula for quantifying the stakes, and the ADL benchmark for specification choice gives applied researchers a practical toolkit for the most consequential decision in observational research: which variables to include as controls.

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A Online Appendix

A.1 Appendix A: Full Cross-Section Derivation

We provide the step-by-step algebra for the cross-section IVB derivation.

Setup. The true model is $y = \beta_0 + \beta_1 D + e$ with $\mathbb{E}[e \mid D] = 0$. The wrong model includes a collider Z: $y = \beta_0^\star + \beta_1^\star D + \beta_2^\star Z + e^\star$. The collider relationship is $Z = \gamma_0 + \gamma_1 D + \gamma_2 y + u$ with $\mathbb{E}[u \mid D, y] = 0$.

Step 1. From the standard OLS formula for a multivariate regression:

$$\beta_1^\star = \frac{\mathrm{Cov}(D, y) \cdot \mathrm{Var}(Z) - \mathrm{Cov}(D, Z) \cdot \mathrm{Cov}(Z, y)}{\mathrm{Var}(D) \cdot \mathrm{Var}(Z) - [\mathrm{Cov}(D, Z)]^2}$$

However, it is more convenient to use the Frisch–Waugh–Lovell approach. By FWL, $\beta_1^\star$ equals the coefficient from regressing y on D after partialling out Z, or equivalently:

$$\beta_1^\star = \frac{\mathrm{Cov}(D, y) - \beta_2^\star \mathrm{Cov}(D, Z)}{\mathrm{Var}(D)}$$

This gives:

$$\beta_1^\star - \beta_1 = -\beta_2^\star \frac{\mathrm{Cov}(D, Z)}{\mathrm{Var}(D)}$$

Step 2. Substitute $y = \beta_0 + \beta_1 D + e$ into $Z = \gamma_0 + \gamma_1 D + \gamma_2 y + u$:

$$\begin{aligned} Z &= \gamma_0 + \gamma_1 D + \gamma_2(\beta_0 + \beta_1 D + e) + u \\ &= (\gamma_0 + \gamma_2 \beta_0) + (\gamma_1 + \gamma_2 \beta_1) D + (\gamma_2 e + u) \end{aligned}$$

Define $\phi_0 = \gamma_0 + \gamma_2 \beta_0$, $\phi_1 = \gamma_1 + \gamma_2 \beta_1$, and $\varepsilon = \gamma_2 e + u$.

Step 3. Since $\mathbb{E}[\varepsilon \mid D] = \gamma_2 \mathbb{E}[e \mid D] + \mathbb{E}[u \mid D] = 0$:

$$\frac{\mathrm{Cov}(D, Z)}{\mathrm{Var}(D)} = \phi_1$$

Step 4. Therefore:

$$\mathrm{IVB} = \beta_1^\star - \beta_1 = -\beta_2^\star \times \phi_1 \qquad \blacksquare$$

A.2 Appendix B: Full FWL/ADL Derivation

We provide the complete derivation for the ADL(1,0) case using FWL.

Setup. The correct model is $y_t = \alpha + \beta D_t + \rho y_{t-1} + e_t$ with $\mathbb{E}[e_t \mid D_t, y_{t-1}] = 0$. The wrong model includes a collider: $y_t = \alpha^\star + \beta^\star D_t + \rho^\star y_{t-1} + \theta^\star Z_t + v_t$.

Step 1 (FWL residualization). Let $W_t = (1, y_{t-1})$. Define residuals:

$$\tilde{y}_t := y_t - \hat{\Pi}_y W_t$$

$$\tilde{D}_t := D_t - \hat{\Pi}_D W_t$$

$$\tilde{Z}_t := Z_t - \hat{\Pi}_Z W_t$$

By FWL, the long regression of y_t on D_t, y_{t-1}, and Z_t is equivalent to the regression of $\tilde{y}_t$ on $\tilde{D}_t$ and $\tilde{Z}_t$.

Step 2 (Two-variable regression formula). In the residualized regression:

$$\beta^\star = \frac{\mathrm{Cov}(\tilde{D}_t, \tilde{y}_t) - \theta^\star \mathrm{Cov}(\tilde{D}_t, \tilde{Z}_t)}{\mathrm{Var}(\tilde{D}_t)}$$

The short model (by FWL applied to the regression of y_t on D_t and y_{t-1}) gives:

$$\beta = \frac{\mathrm{Cov}(\tilde{D}_t, \tilde{y}_t)}{\mathrm{Var}(\tilde{D}_t)}$$

Step 3 (Bias). Subtracting:

$$\beta^\star - \beta = -\theta^\star \frac{\mathrm{Cov}(\tilde{D}_t, \tilde{Z}_t)}{\mathrm{Var}(\tilde{D}_t)}$$

Step 4 (Auxiliary regression). Consider $Z_t = a + \pi D_t + \lambda y_{t-1} + \eta_t$. By FWL, π equals the coefficient from regressing $\tilde{Z}_t$ on $\tilde{D}_t$:

$$\pi = \frac{\mathrm{Cov}(\tilde{D}_t, \tilde{Z}_t)}{\mathrm{Var}(\tilde{D}_t)}$$

Step 5 (Result).

$$\mathrm{IVB}(\beta) = \beta^\star - \beta = -\theta^\star \times \pi \qquad \blacksquare$$

ADL(p,q) extension. For general ADL(p,q), the legitimate control set becomes $W_t = (1, y_{t-1}, \ldots, y_{t-p}, D_{t-1}, \ldots, D_{t-q})$ (excluding D_{t-j} when computing the bias for β_j). The FWL argument carries through identically, yielding $\mathrm{IVB}(\beta_j) = -\theta^\star \times \pi_j$ where π_j is the coefficient on D_{t-j} in the auxiliary regression $Z_t = a + \pi_j D_{t-j} + W_t'\delta + \eta_t$.

■

A.3 Appendix C: Interaction with Nickell Bias

In panels with unit fixed effects and short time dimensions, the OLS estimates of ADL coefficients suffer from Nickell bias (Nickell, 1981), which arises because the within-transformation creates a mechanical correlation between the demeaned lagged dependent variable $\tilde{y}_{i,t-1} = y_{i,t-1} - \bar{y}_i$ and the demeaned error term $\tilde{e}_{it} = e_{it} - \bar{e}_i$. The bias is of order $O(1/T)$: it can be severe when $T < 10$ but becomes negligible as T grows (Judson and Owen, 1999).

Because the IVB formula is an algebraic identity—it decomposes the finite-sample difference between two nested OLS regressions—it remains exactly valid even in the presence of Nickell bias. The identity $\hat{\beta}^\star - \hat{\beta} = -\hat{\theta}^\star \times \hat{\pi}$ holds by construction. However, the *causal interpretation* of the components changes: in a panel with short T, both $\hat{\theta}^\star$ and $\hat{\pi}$ reflect a mixture of the true collider relationship and the dynamic bias introduced by the within-transformation. The net effect on the IVB product is ambiguous in sign but bounded by $O(1/T)$, and simulation evidence in Judson and Owen (1999) suggests it becomes negligible once T exceeds 20–30 periods. The *direction* of the IVB tends to be preserved even when magnitudes are contaminated: the sign of $-\hat{\theta}^\star\hat{\pi}$ remains informative about the direction of collider bias. In short panels ($T < 20$), bias-corrected estimators (Arellano and Bond, 1991) or the half-panel jackknife may be needed before giving the IVB components a purely collider-bias interpretation.

A.4 Appendix D: Lagging the Treatment Is Not a Mechanical Cure

Applied work sometimes replaces a contemporaneous treatment D_t with a longer lag (e.g., D_{t-3}) under the intuition that this should make collider bias less likely. The IVB decomposition clarifies that this is not true in general.

Proposition 4 (Lag Substitution Without Lagged Outcome)**.** *Consider a model without lagged dependent variables. For any target lag $k \geq 0$, define $y_t = \alpha + \beta_k D_{t-k} + W_t'\gamma + e_t$ as the correct (short) model and $y_t = \alpha^\star + \beta_k^\star D_{t-k} + W_t'\gamma^\star + \theta^\star Z_t + v_t$ as the long model. Then $\beta_k^\star - \beta_k = -\theta^\star \pi_k$, where π_k is the coefficient on D_{t-k} in $Z_t = a + \pi_k D_{t-k} + W_t'\delta + \eta_t$.*

Now suppose the collider process is $Z_t = a + \pi_0 D_t + \pi_1 D_{t-1} + W_t'\delta + \eta_t$. Let $\widetilde{X}$ denote residualization of X on W_t, and define $\widetilde{D}_{t-\ell} = \omega_{\ell k}\widetilde{D}_{t-k} + r_{\ell t}$ with $\mathrm{Cov}(\widetilde{D}_{t-k}, r_{\ell t}) = 0$. Then $\pi_k = \pi_0\omega_{0k} + \pi_1\omega_{1k}$, so that

$$\beta_k^\star - \beta_k = -\theta^\star \left(\pi_0\omega_{0k} + \pi_1\omega_{1k}\right). \tag{16}$$

This shows that moving from D_{t-1} to D_{t-3} does not mechanically eliminate IVB. More generally, for any number of lags L: $\pi_k = \sum_{\ell=0}^{L} \pi_\ell \omega_{\ell k}$, so longer lags can attenuate IVB when the $\omega_{\ell k}$ terms are small but do not guarantee zero collider bias.

Corollary 2 (IVB Decay Under AR(1) Treatment)**.** *Under the conditions of Proposition 4, suppose further that (i) the demeaned treatment follows a stationary AR(1) process with $|\rho_D| < 1$, and (ii) the collider depends only on the contemporaneous treatment. Then $IVB(k) \approx IVB(0) \cdot \rho_D^k$.*

For highly persistent treatments ($\rho_D \approx 0.95$), the IVB decays slowly: IVB(5) $\approx 0.77 \times$ IVB(0). For moderate autocorrelation ($\rho_D \approx 0.5$), decay is rapid: IVB(3) $\approx 0.13 \times$ IVB(0). A useful rule of thumb is that $k^\star \approx \log(\text{SE}/|\text{IVB}(0)|)/\log(\rho_D)$ lags are needed for IVB to fall below one standard error.

A.5 Appendix E: Simulation Code Description

All simulations reported in Section~5 are implemented in R using the `fixest` and `data.table` packages. Simulation scripts are in the `simulations/` directory; the replication code for Section~6 applications is embedded in the R Markdown source file. All random number generators are seeded for reproducibility. Each simulation uses 500 replications and a 100-period burn-in to ensure stationarity.

DGP 1 (Collider/Dual-Role, Section~5.1): $N = 100$ units, $T = 30$ periods. Treatment is binary with staggered adoption ($D_{it} = \mathbf{1}(t \geq T_i^\star)$). The outcome follows $Y_{it} = \alpha_i^Y + \beta D_{it} + \gamma_Y Z_{i,t-1} + \rho_Y Y_{i,t-1} + e_{it}$ with $\beta = 1$, $\rho_Y = 0.5$, $\gamma_Y = 0.2$. The covariate Z is dual-role: $Z_{it} = \alpha_i^Z + \delta_D D_{it} + \delta_Y Y_{it} + \rho_Z Z_{i,t-1} + \nu_{it}$, so Z_t is a collider ($D_t \to Z_t \leftarrow Y_t$) and Z_{t-1} is a confounder ($Z_{t-1} \to Y_t$). Grid: prob_switch $\times\, \delta_D \times \delta_Y \times \rho_Z = 4 \times 2 \times 3 \times 2 = 48$ scenarios. Nine estimation strategies are compared, from TWFE without controls to ADL with all lags. Script: `simulations/v4_mechanisms/sim_mechC_adl.R`.

DGP 2 (Contemporaneous Mediator, Section~5.2): $N = 100$, $T = 30$. Treatment is continuous. Z_t is a pure contemporaneous mediator: $Z_{it} = \alpha_i^Z + \delta_D D_{it} + f_{\text{nl}}(D_{it}) + \rho_Z Z_{i,t-1} + \nu_{it}$ (no $Y_t \to Z_t$ channel). The outcome includes the mediator: $Y_{it} = \alpha_i^Y + \beta D_{it} + \theta Z_{it} + \rho_Y Y_{i,t-1} + e_{it}$, with $\beta = 1$, $\theta = 0.5$, $\delta_D = 0.4$, yielding a true total effect of 1.2 under linearity. A second variant adds confounding through Z_{t-1}: $\gamma_D = 0.15$, $\gamma_Y = 0.2$. Grid: 2 variants $\times$ (1 baseline + 8 nonlinearity types) $\times$ 2 persistence levels. Scripts: `simulations/overcontrol/sim_overcontrol.R`, `simulations/nonlinearity/sim_overcontrol_contemporaneous.R`.

DGP 3 (Nonlinearity, Section~5.3): Same dual-role structure as DGP~1 but with continuous treatment and nonlinear collider channels. Eight nonlinearity types are tested: six bounded ($\log(1 + D^2)$, $\log(1 + D^4)$, soft-clamped polynomial, $\sin(D)$, inverse logit, tanh) and two unbounded ($\text{sign}(D)|D|^{1.5}$, $D \log(1 + |D|)$).

All are calibrated so that the nonlinear contribution at $D = \mathrm{SD}(D_{\text{within}})$ equals strength $\times\, \delta_D \times \mathrm{SD}(D_{\text{within}})$. Grid: 84 scenarios. Script: `simulations/nonlinearity/sim_nl_collider.R`.

DGP 4 (Feedback, Section~5.3): Same dual-role structure as DGP~1 but with continuous treatment and outcome-to-treatment feedback: $D_{it} = \alpha_i^D + \phi Y_{i,t-1} + \gamma_D Z_{i,t-1} + \rho_D D_{i,t-1} + u_{it}$, where $\phi \in \{0, 0.05, 0.1, 0.15\}$ violates strict exogeneity. Grid: $\phi \times \rho_Z = 4 \times 2 = 8$ scenarios. Script: `simulations/dynamics/sim_feedback_Y_to_D.R`.

A.6 Appendix F: IVB in the Potential Outcomes Framework

This appendix re-derives the IVB formula in the potential outcomes notation standard in the DID literature, and connects it to the decomposition in Caetano and Callaway (2024).

Setup. Consider two periods: $t^* - 1$ (pre-treatment) and t^* (post-treatment). Treatment is binary: $D_i \in \{0, 1\}$ (no unit treated at $t^* - 1$). Each unit has potential outcomes $Y_{it}(1)$, $Y_{it}(0)$ and potential covariates $Z_{it}(1)$, $Z_{it}(0)$. The estimand is $ATT = \mathbb{E}[Y_{t^*}(1) - Y_{t^*}(0) \mid D = 1]$.

First-differenced TWFE. With two periods, the TWFE specification $Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \beta Z_{it} + e_{it}$ reduces to:

$$\Delta Y_{it^*} = \tau D_i + \beta \Delta Z_{it^*} + \Delta e_{it^*}$$

where Δ denotes first differences. The short model (without Z) gives τ_S.

IVB derivation. By FWL applied to this first-differenced regression:

$$\tau - \tau_S = -\beta \times \pi, \qquad \pi = \frac{\mathrm{Cov}(D, \Delta Z)}{\mathrm{Var}(D)}$$

This is the IVB formula in potential outcomes notation. Both β and π are directly estimable. The component π captures how much the treatment changes the covariate: if D does not affect Z (i.e., $Z_{it^*}(1) = Z_{it^*}(0)$ for all i), then $\pi = 0$ and including Z is harmless. When D affects Z, $\pi \neq 0$ and the IVB is non-zero.

Connection to Caetano and Callaway (2024). Under their Assumption~3 (strict exogeneity of covariates), D does not affect Z, so $\pi = 0$ and the IVB does not appear in their decomposition. Their Proposition~1 decomposes TWFE bias into terms (A)–(C) reflecting hidden linearity bias. When D does affect Z—violating their Assumption~3, which is the "bad controls'' case—the IVB adds to their decomposition:

$$\hat{\tau}_{\text{TWFE}} = ATT + \underbrace{(-\beta \times \pi)}_{\text{IVB (bad control bias)}} + \underbrace{\text{Terms (A)–(C)}}_{\text{hidden linearity bias}}$$

The IVB formula quantifies the first component; the diagnostics of Caetano and Callaway (2024) address the second. The two are complementary.

Why TSCS has a structural advantage. In the two-period DID, the researcher has only Z_{t^*} (contemporaneous). If Z is both a confounder (parallel trends hold only conditional on Z) and a mediator ($D \to Z \to Y$), an inescapable dilemma arises: including Z_{t^*} satisfies conditional parallel trends but generates over-control; excluding it avoids over-control but violates conditional parallel trends. This is the problem motivating the doubly robust and imputation-based methods of Caetano et al. (2022).

In TSCS ($T > 2$), the temporal structure resolves this dilemma. The ADL specification conditions on Z_{t-1}, which is pre-treatment for the contemporaneous effect of D_t on Y_t. Under sequential ignorability (Blackwell and Glynn, 2018), conditioning on $\{Y_{t-1}, Z_{t-1}\}$ blocks all backdoor paths—including those through colliders—without blocking the contemporaneous mediator channel $D_t \to Z_t \to Y_t$. The lagged value absorbs confounding (satisfying conditional parallel trends) while the contemporaneous value is left out of the regression, preserving the total effect. This separation is the timing logic that the ADL benchmark in Section~5.4 exploits.

A.7 Appendix G: Variance of the IVB Estimator

The IVB diagnostic $\widehat{\text{IVB}} = -\hat{\theta}^\star \hat{\pi}$ is a product of two estimators. Its sampling variance can be approximated via the delta method. Write $\text{IVB} = -\theta^\star \pi$ and let $\hat{\theta}^\star$ and $\hat{\pi}$ denote the OLS estimators from the long model and the auxiliary regression, respectively. By the delta method:

$$\text{Var}(\widehat{\text{IVB}}) \approx \pi^2 \,\text{Var}(\hat{\theta}^\star) + \theta^{\star 2} \,\text{Var}(\hat{\pi}) + 2\theta^\star \pi \,\text{Cov}(\hat{\theta}^\star, \hat{\pi}). \tag{17}$$

The first two terms on the right-hand side are readily available from the standard output of the long regression ($\text{Var}(\hat{\theta}^\star)$) and the auxiliary regression ($\text{Var}(\hat{\pi})$). The cross-covariance term $\text{Cov}(\hat{\theta}^\star, \hat{\pi})$ is not standard, since the two estimators come from different regressions. It can be estimated via seemingly unrelated regressions (SUR) or, more simply, via cluster bootstrap.

For a conservative bound that avoids estimating the cross-covariance, apply the Cauchy–Schwarz inequality:

$$\text{SE}(\widehat{\text{IVB}}) \leq |\hat{\pi}| \cdot \text{SE}(\hat{\theta}^\star) + |\hat{\theta}^\star| \cdot \text{SE}(\hat{\pi}). \tag{18}$$

This upper bound is computable from standard regression output alone, without any additional estimation. If the bound is small relative to $\text{SE}(\hat{\beta})$, the researcher can conclude that the IVB is precisely estimated as

small (or large). If the bound is inconclusive—i.e., the upper bound on SE($\widehat{\text{IVB}}$) is of similar magnitude to $|\widehat{\text{IVB}}|$ itself—then a cluster bootstrap is recommended to obtain a tighter interval.

We recommend the following two-step procedure: (1) compute the upper bound in Equation~(18) as a quick screening tool; (2) if the bound is inconclusive, run a cluster bootstrap that jointly resamples both regressions to obtain SE($\widehat{\text{IVB}}$) and a confidence interval directly.

A.8 Appendix H: Control Classification and Full IVB Results

Following the diagnostic recipe in Section~4.6 (Step~0), we classify each control variable in the six replicated studies according to its causal role relative to the treatment (D) and outcome (Y). A control is classified as a **collider** when credible evidence from separate studies supports both $D \to Z$ and $Y \to Z$; as a **confounder** when the evidence supports $Z \to D$ and/or $Z \to Y$ (but not $D \to Z$); as a **mediator** when $D \to Z \to Y$; and as **predetermined/neutral** when no contemporaneous causal pathway is plausible (e.g., geographic or demographic variables, or lagged variables where the lag structure blocks the collider channel). Variables with evidence supporting both confounding and collider channels are labeled **collider + confounder**. The classification criteria are consistent with the sufficient conditions stated in Section~4.7.

A.8.1 Claassen (2020) — D: public support, Y: liberal democracy

Table A1: Control classification for Claassen (2020).

Control	$D \to Z$	$Y \to Z$	Classification	Key references
Log GDP p.c.	Weak (institutional)	Democracy → GDP	Collider (weak)	Acemoglu et al. (2019)
GDP growth	Same channel	Same channel	Collider (weak)	Acemoglu et al. (2019)
Regional democracy	Diffusion	Composition	Predetermined	—
% Muslim	None	None (absorbed by FE)	Predetermined	—
Resource dep.	None	Resource curse is $Z \to Y$	Confounder	Ross (2001)

Collider candidate for Table~2: Log GDP p.c. GDP growth shares the same mechanism but is excluded from the main table to avoid double-counting the same causal channel. Both GDP variables are classified as "Collider (weak)'' because the $D \to Z$ channel (public support → GDP) is indirect and operates through institutional quality rather than a direct causal mechanism.

A.8.2 Leipziger (2024) — D: binary democracy, Y: ethnic inequality (SEI)

Table A2: Control classification for Leipziger (2024).

Control	$D \to Z$	$Y \to Z$	Classification	Key references
Log GDP p.c.	Democracy → GDP	Ethnic ineq. → GDP	**Collider**	Acemoglu et al. (2019); Grundler and Link (2024)
Oil income p.c.	None	Geological	Predetermined	—
Civil war	Democracy → civil war	Ethnic ineq. → civil war	**Collider**	Cederman, Gleditsch and Buhaug (2013)
Ethnic frac.	None	Demographic	Predetermined	—
GDP growth	Same channel as GDP p.c.	Same channel	**Collider**	Acemoglu et al. (2019)

Collider candidates: Log GDP p.c. (baseline spec.), Civil war and GDP growth (extended spec.).

A.8.3 Blair et al. (2022) — D: UN peacekeeping, Y: democratization

Table A3: Control classification for Blair et al. (2022).

Control	$D \to Z$	$Y \to Z$	Classification	Key references
Population	Demographic	Demographic	Predetermined	—
Foreign Aid	PKO → bundled aid	Democracy → aid	**Collider**	Bermeo (2016)
GDP p.c.	PKO → stability → GDP	Democracy → GDP	**Collider**	Acemoglu et al. (2019)
Refugees/IDPs	PKO → fewer displaced	Democracy → less repression	**Collider + Confounder**	Costalli, Di Salvatore and Ruggeri (2024); Moore and Shellman (2004)
Literacy	Education stock (decades)	Weak	Predetermined	—
Fuel exports	Geological	Geological	Predetermined	—

Collider candidates: Foreign Aid, GDP per capita, Refugees/IDPs. Refugees/IDPs is classified as collider + confounder: PKO mandates reduce displacement ($D \to Z$; Costalli, Di Salvatore and Ruggeri, 2024) and democratization reduces refugee-generating repression ($Y \to Z$; Moore and Shellman, 2004), but refugee

crises also trigger PKO deployment ($Z \to D$), making the variable simultaneously a confounder (as with GDP per capita in Rogowski et al. 2022).

A.8.4 Albers et al. (2023) — D: government turnover, Y: tax collection

Table A4: Control classification for Albers et al. (2023).

Control	$D \to Z$	$Y \to Z$	Classification	Key references
Independence	Historical	Structural	Predetermined	—
Hyperinflation	Turnover → fiscal → inflation	Tax → fiscal balance	**Collider**	Sargent and Wallace (1981)
Civil war (lag)	Lag protects	Lag protects	Predetermined	—
GDP growth	Turnover → uncertainty → GDP	Tax → investment → GDP	**Collider**	—
Drought (lag)	Exogenous	Exogenous	Predetermined	—
Liberal democracy	Turnover → regime openness	Fiscal bargaining	**Collider (weak)**	Besley and Persson (2011)
Secession	Ethnic-political	Structural	Predetermined	—
Socialist	Ideological	Structural	Predetermined	—
Alliance (G5)	Weak	Weak	Predetermined	—
Market access	Weak	Creditworthiness	Ambiguous	—
Ext. default	Weak	Fiscal capacity	Ambiguous	—
Intl. war (lag)	Geopolitical	Geopolitical	Predetermined	—

Collider candidates: Hyperinflation, GDP growth (YoY), Liberal democracy. Note: standard errors for Albers et al. are not available from the published replication materials, so IVB/SE cannot be computed.

A.8.5 Rogowski et al. (2022) — D: log post offices, Y: GDP growth

Table A5: Control classification for Rogowski et al. (2022).

Control	$D \to Z$	$Y \to Z$	Classification	Key references
Log GDP p.c.	PO → GDP (paper's mechanism)	Growth → levels (identity)	**Collider + Confounder**	Acemoglu et (2019)
Log population	Population → PO (demand)	Pop. → growth (Solow)	Confounder	—
Urbanization	Weak	Growth → urbanization	Weak	—
Polity2	PO → democracy (slow)	Growth → democratization	Weak	—

Collider candidate: Log GDP p.c. (with caveat that it is simultaneously a confounder through conditional convergence; see Section~6.2).

A.8.6 Ballard-Rosa et al. (2021) — D: left government, Y: domestic-currency bonds

Table A6: Control classification for Ballard-Rosa et al. (2021).

Control	$D \to Z$	$Y \to Z$	Classification	Key references
Inflation crisis	Left → inflation	Bonds → inflation expect.	**Collider**	Hibbs (Alesina (1987)
IMF program	Left → fiscal policy → IMF	Bonds → fiscal crisis → IMF	**Collider**	Vreeland (Eichengreen, mann and Pa (2005)
Sov. debt crisis	Left → fiscal expansion → crisis	Bonds → currency mismatch → crisis	**Collider**	Alesina and Tal (1990); Eichen Hausmann Panizza (2005)
Log GDP p.c.	Left → GDP (weak, lagged)	Weak	Predetermined	—
GDP growth	Weak	Weak	Predetermined	—
Ext. debt/GDP	Weak	Weak	Predetermined	—
Curr. acct/GDP	Weak	Weak	Predetermined	—
Trade/GDP	Structural	Structural	Predetermined	—
Oil rents/GDP	Geological	Geological	Predetermined	—
FDI/GDP	Weak	Weak	Predetermined	—
US Treasury 10yr	Exogenous	Exogenous	Predetermined	—
Pegged XR	Structural	Weak	Predetermined	—
High CBI	Institutional	Weak	Predetermined	—
Chinn-Ito open.	Structural	Weak	Predetermined	—
Currency crisis	Weak	Weak	Predetermined	—
Democracy (V-Dem)	Weak	Weak	Predetermined	—

Collider candidates: Inflation crisis, IMF program, Sovereign debt crisis. All three share a common mechanism: left-government fiscal policies create fiscal stress ($D \to Z$; Alesina and Tabellini 1990; Alesina 1987), while the currency denomination of sovereign bonds affects crisis vulnerability through balance-sheet mismatches ($Y \to Z$; Eichengreen, Hausmann and Panizza 2005). For IMF programs specifically, Vreeland

(2003) shows that domestic political factors—including government ideology—drive program participation beyond economic necessity.

A.8.7 Full Mechanical IVB Decomposition

Table~A7 reports the complete IVB decomposition for all 55 study-control combinations, regardless of collider plausibility. This table is provided for completeness and transparency; the substantive interpretation of any individual IVB as collider bias requires the causal reasoning documented in the classification tables above.

Table A7: Full mechanical IVB decomposition. All controls are included regardless of collider plausibility. IVB/$\hat{\beta}$ is reported only when the treatment effect is significant ($|t| \geq 1.96$); "—" indicates $\hat{\beta} \approx 0$ or SE unavailable.

Study	Control	IVB	|IVB/SE|	IVB/$\hat{\beta}$ (%)
Claassen (OLS)	Log GDP p.c.	-0.0010	-0.01	-0.4%
Claassen (OLS)	GDP growth	-0.0009	-0.01	-0.3%
Claassen (OLS)	Regional democracy	-0.0092	-0.11	-3.4%
Claassen (OLS)	% Muslim	0.0010	0.01	0.4%
Claassen (OLS)	Resource dep.	0.0083	0.10	3.1%
Claassen (FE)	Log GDP p.c.	-0.0165	-0.08	—
Claassen (FE)	GDP growth	0.0123	0.06	—
Claassen (FE)	Regional democracy	-0.0020	-0.01	—
Claassen (FE)	Resource dep.	-0.0001	-0.00	—
Leipziger (SEI)	Log GDP p.c.	0.0056	0.48	-15.9%
Leipziger (GRG)	Log GDP p.c.	-0.0003	-0.02	0.9%
Leipziger (GGINI)	Log GDP p.c.	-0.0008	-0.12	—
Leipziger (SEI ext.)	Log GDP p.c.	0.0048	0.41	-15.4%
Leipziger (SEI ext.)	Oil income p.c.	-0.0000	-0.00	0.1%
Leipziger (SEI ext.)	Civil war	0.0002	0.02	-0.7%
Leipziger (SEI ext.)	Ethnic frac.	0.0007	0.06	-2.1%
Leipziger (SEI ext.)	GDP growth	-0.0000	-0.00	0.0%
Blair et al.	Population	-0.0005	-0.01	-0.5%
Blair et al.	Foreign Aid	0.0029	0.07	2.4%
Blair et al.	GDP per capita	0.0055	0.14	4.7%
Blair et al.	Refugees/IDPs	0.0050	0.13	4.2%
Blair et al.	Literacy	0.0000	0.00	0.0%

Table A7: Full mechanical IVB decomposition. All controls are included regardless of collider plausibility. IVB/$\hat{\beta}$ is reported only when the treatment effect is significant ($|t| \geq 1.96$); "—" indicates $\hat{\beta} \approx 0$ or SE unavailable. *(continued)*

Study	Control	IVB	|IVB/SE|	IVB/$\hat{\beta}$ (%)
Blair et al.	Fuel exports	-0.0010	-0.03	-0.8%
Albers et al.	Independence	-0.0409	—	—
Albers et al.	Hyperinflation	0.0288	—	—
Albers et al.	Civil war (PRIO)	0.0210	—	—
Albers et al.	GDP growth (YoY)	0.0163	—	—
Albers et al.	Drought	-0.0150	—	—
Albers et al.	Liberal democracy	-0.0131	—	—
Albers et al.	Secession	0.0124	—	—
Albers et al.	Socialist	0.0070	—	—
Albers et al.	Alliance (G5)	0.0061	—	—
Albers et al.	Market access	0.0055	—	—
Albers et al.	External default	-0.0014	—	—
Albers et al.	Intl war (PRIO)	-0.0007	—	—
Rogowski et al.	Log GDP p.c.	0.0115	2.11	58.0%
Rogowski et al.	Log population	-0.0008	-0.15	-4.0%
Rogowski et al.	Urbanization	0.0006	0.11	3.0%
Rogowski et al.	Polity2	0.0008	0.14	3.8%
Ballard-Rosa et al.	Log GDP p.c.	-0.0057	-0.13	-5.2%
Ballard-Rosa et al.	GDP growth	0.0003	0.01	0.3%
Ballard-Rosa et al.	Ext. debt/GDP	0.0057	0.13	5.2%
Ballard-Rosa et al.	Curr. acct/GDP	-0.0018	-0.04	-1.7%
Ballard-Rosa et al.	Trade/GDP	0.0024	0.05	2.2%
Ballard-Rosa et al.	Oil rents/GDP	0.0002	0.00	0.2%
Ballard-Rosa et al.	FDI/GDP	-0.0007	-0.02	-0.6%
Ballard-Rosa et al.	US Treasury 10yr	0.0012	0.03	1.1%
Ballard-Rosa et al.	Pegged XR	-0.0001	-0.00	-0.1%
Ballard-Rosa et al.	High CBI	-0.0005	-0.01	-0.4%
Ballard-Rosa et al.	Chinn-Ito open.	0.0010	0.02	0.9%
Ballard-Rosa et al.	IMF program	-0.0086	-0.19	-7.9%

Table A7: Full mechanical IVB decomposition. All controls are included regardless of collider plausibility. IVB/$\hat{\beta}$ is reported only when the treatment effect is significant ($|t| \geq 1.96$); "—" indicates $\hat{\beta} \approx 0$ or SE unavailable. *(continued)*

Study	Control	IVB	|IVB/SE|	IVB/$\hat{\beta}$ (%)
Ballard-Rosa et al.	Currency crisis	0.0006	0.01	0.6%
Ballard-Rosa et al.	Inflation crisis	-0.0125	-0.28	-11.4%
Ballard-Rosa et al.	Sov. debt crisis	-0.0038	-0.09	-3.5%
Ballard-Rosa et al.	Democracy (V-Dem)	0.0004	0.01	0.3%